

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Academic Year: 2019 - 2020 (Odd Semester)

Degree, Semester & Branch : V Semester B.E (Common to ECE, EEE, MECH)
Course(Subject) Code & Title : OIT551 Database Management Systems
Name of the Faculty member : Mr. K.Vignesh Saravanan, AP / CSE

Date: 11.07.2019

Innovative Practice: JIGSAW
Topics: Modelling of ER diagram (any 7 applications)

Objectives:

- **O1:** To encourage the students to think, identify and design ER diagrams.
- **O2:** To improve the listening, communication, creativity skills of the students.

Justification for choosing this topic:

Entity Relationship diagram is used to define the data elements and relationship for a specified system. It develops a conceptual design for the database. Different applications are assigned to students under each group. Students are provided the opportunity to excel themselves in a particular application and share that knowledge with their team members. The prime objective of JIGSAW is to promote both self and peer teaching which requires students to understand the concept very clearly and engage in discussion and learning.

Group activity on modeling the ER concepts

To understand the modelling of any application system, the students are made to work into groups to discuss and understand the ER models. The students are asked to identify different applications in various domains. Thus they have listed out the different applications below:

1. Library Management system
2. Banking system
3. Hospital Management system
4. Employee payroll system
5. College management system
6. Hotel management system
7. Airline Reservation system

Students were divided into **7 groups with 7 members in each group**. Each student in a group is assigned with any one application. The course instructor delivered the instructions how to model the ER diagram.

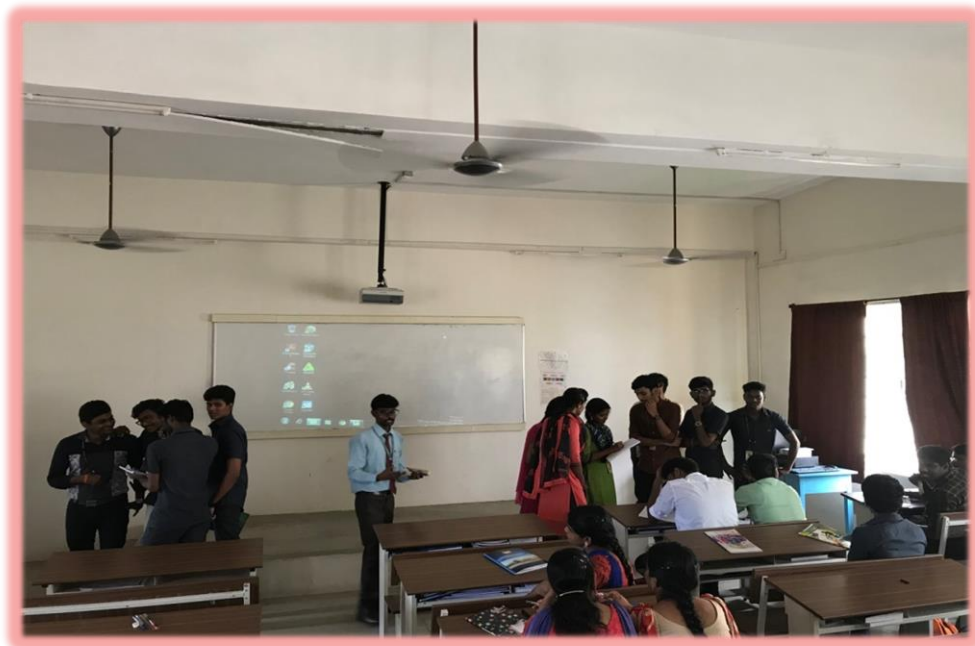
For collaborative teamwork, the student who are assigned with the same applications in each group are picked out and framed as a team – “**Expert Group**”. Thus for each application there will be an expert group. The expert group members discuss within themselves and model the ER diagram for their application.

After they have modelled the application, the expert group is collapsed and all the students are re-grouped into their original group (initial team). Now each member in the team will be an expert in any one application. They have to share their ideas to all the other members in the group thus enabling each student to learn all the applications. This will enable the students to get insight knowledge in this topic.

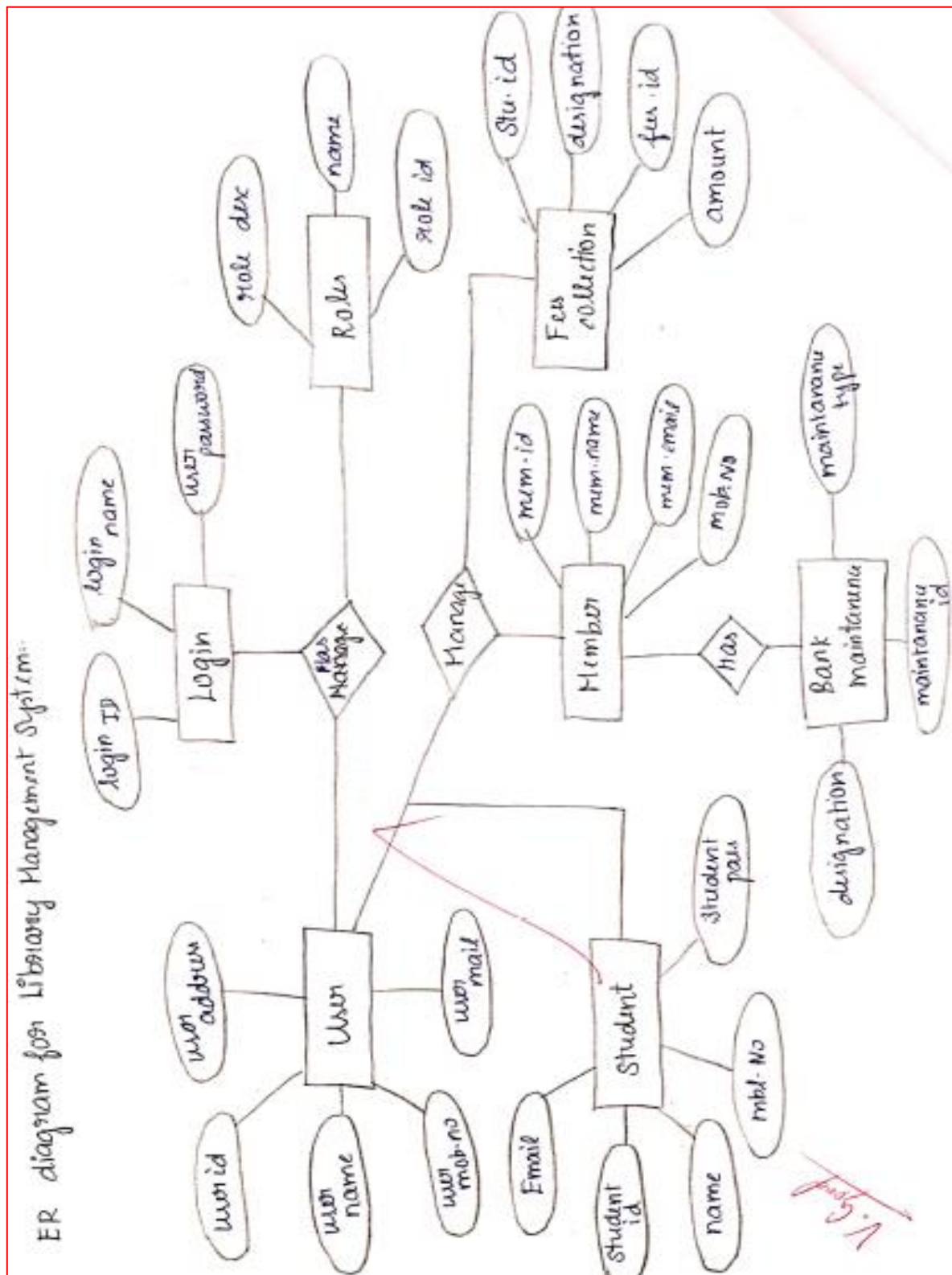
OUTCOME:

Through this activity the students are **made to work in a team and share their ideas** to others and also **improve their communication skills**. Thus the students were made to understand that the team work will give a better solution. The **students are exposed how to identify and design a specific application**, which will help them to do the same when they work in an organization.

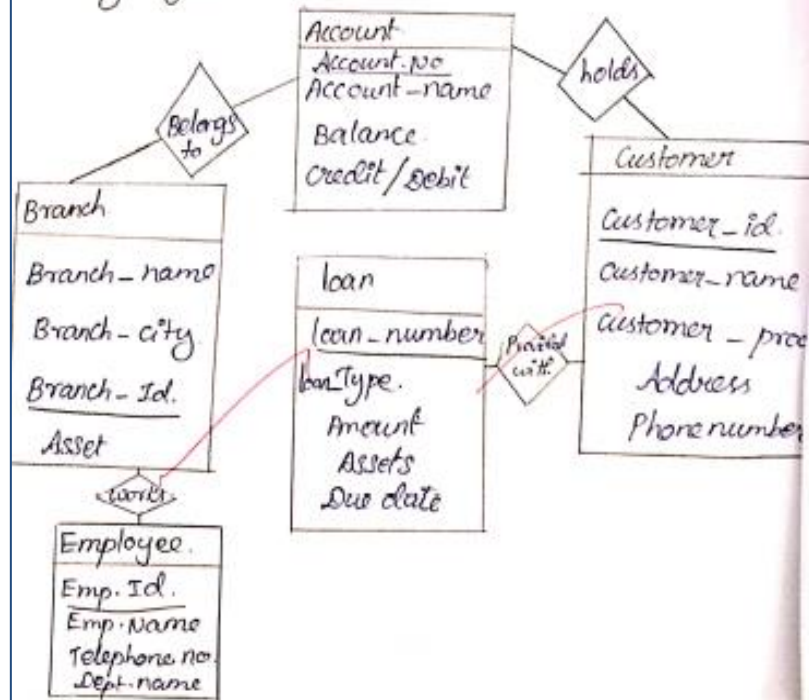
Glimpse of Students working as a team



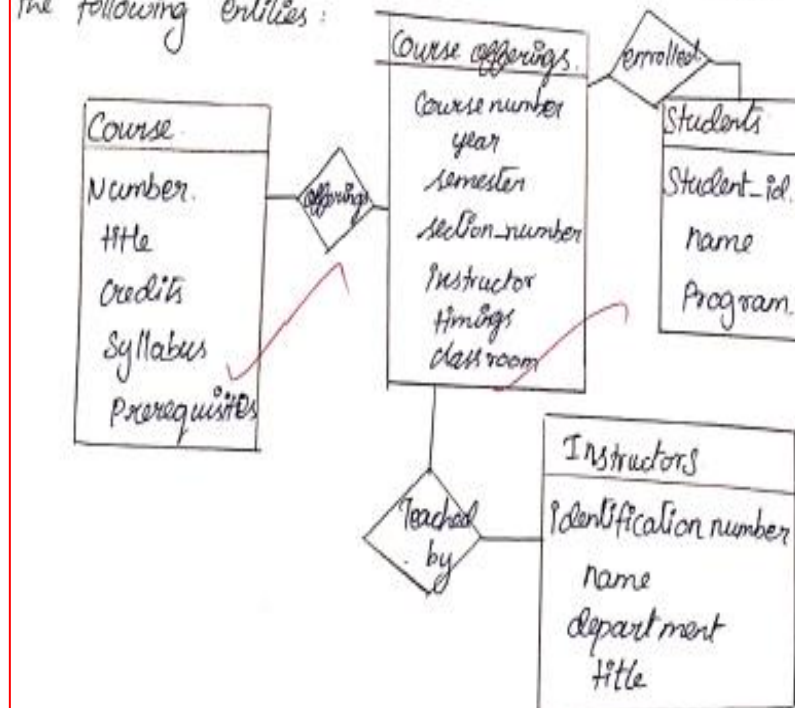
ER Models designed by the students



Banking Systems:



A university registrar's office maintains data about the following entities:



Relevance Mapping - This activity helps in attaining the following Objectives – PO mapping:

Course Outcome / Programme Outcome	PO1	PO2	PO3	PO9	PO10	PO12
O1 - To encourage the students to think, identify and design ER diagrams	3	3	3	-	-	1
O2 - To improve the listening, communication, creativity skills of the students	-	-	-	3	3	2

Reflection Critique

Challenges:

- The grouping of the students was difficult, since students are from various departments (ECE-19, EEE-16, MECH-20). The team size of 7 members were planned with 7 teams, so remaining 1 team with 6 members.
- Students felt uncomfortable to work with other department students, so grouping has to be done with same department.
- Some students in the team did not participate actively in the team.
- Individual assessment/observation were difficult.

Steps taken for address the challenges:

- Initially, I started to group as per planned with 7 teams, but later grouped the members with same department, so it was time-consuming.
- I motivated the students who did not involve in activity and encouraged that the team as well as individual members will get any small prizes.
- To observe the understanding of each student, at the end of the session, I asked any two team (all members) to come front and discuss the entity diagram to all.